

Incentive

$$\nabla F = \frac{J}{c\epsilon_0}$$

I_m = Design TR with no sources

$$\nabla F = 0$$

↓ greens

Convolutions $\rightarrow$ $F(\vec{x}) = \frac{1}{c i} \int \frac{F(\vec{x}')}{|\vec{x}' - \vec{x}|} d\vec{x}'$

on de
multiplication

$$J = c p - \vec{J}$$

multiplication

$$\nabla = \left(\frac{1}{c} \frac{\partial}{\partial t} + \vec{\nabla} \right)$$

multiplication

$$F = \vec{E} + \underbrace{ic \vec{B}}_{\text{divergence}}$$

$$\left(\frac{1}{c} \frac{\partial}{\partial t} + \vec{\nabla} \right) (\vec{E} + ic \vec{B}) = \frac{c p - \vec{J}}{c \epsilon_0}$$

$$\frac{1}{c} \frac{\partial \vec{E}}{\partial t} + i \frac{\partial \vec{B}}{\partial t} + \vec{\nabla} \vec{E} + ic \vec{\nabla} \vec{B} = \frac{p}{\epsilon_0} - \frac{\vec{J}}{c \epsilon_0}$$

$$\vec{\nabla} \cdot \vec{E} + \vec{\nabla} \wedge \vec{E}$$

$$i c \vec{\nabla} \cdot \vec{B} + i c \vec{\nabla} \vee \vec{B}$$

$$\underbrace{\vec{\nabla} \cdot \vec{E}}_{\text{Scalar}} + \underbrace{\frac{1}{c} \frac{\partial \vec{E}}{\partial t}}_{\text{Vector}} + \underbrace{i c \vec{\nabla} \vee \vec{B}}_{\text{Vector}} + \underbrace{i \frac{\partial \vec{B}}{\partial t} + \vec{\nabla} \wedge \vec{E}}_{\text{Bivector}} + \underbrace{i c \vec{\nabla} \cdot \vec{B}}_{\text{Trivector}} = \underbrace{c \rho - \vec{J}}_{\text{Scalar, Vector}}$$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$\frac{1}{c} \frac{\partial \vec{E}}{\partial t} + i c \vec{\nabla} \vee \vec{B} = \frac{\vec{J}}{c \epsilon_0}$$

$$\frac{1}{c^2} \frac{\partial \vec{E}}{\partial t} + i \vec{\nabla} \vee \vec{B} = \frac{\vec{J}}{c \epsilon_0}$$

$$\mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} - \vec{\nabla} \times \vec{B} = -\mu_0 \vec{J}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

ampere

in $\mathbb{R}^2$

Vectors $\rightarrow \vec{u} = a\hat{x} + b\hat{y}$

in Geometric Algebra $V = a + b\hat{x} + c\hat{y} + d\hat{x}\hat{y}$

$\nearrow$ multivector

$\downarrow$ scalar

$\underbrace{\hspace{2cm}}$ vectors

$\underbrace{\hspace{2cm}}$ bivector

Bivector in 2D $\hat{x}\hat{y} = i \rightarrow$ pseudo scalar

$$\vec{v} = 2\hat{x} + 3\hat{y}$$

$$\hat{x}\hat{x} = 1$$

$$\begin{aligned}\vec{v} \cdot \hat{x}\hat{y} &= 2\hat{x}\hat{x}\hat{y} + 3\hat{y}\hat{x}\hat{y} \quad \text{anticommuting,} \\ &= 2\hat{y} - 3\hat{x}\end{aligned}$$

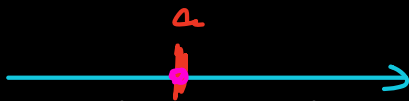
Observe that $i^2 = \hat{x}\hat{y}\hat{x}\hat{y}$

$$i^2 = -\hat{x}\hat{x}\hat{y}\hat{y}$$

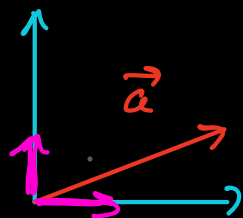
$$i^2 = -1$$

In 2D i is a Bivector

Scalar



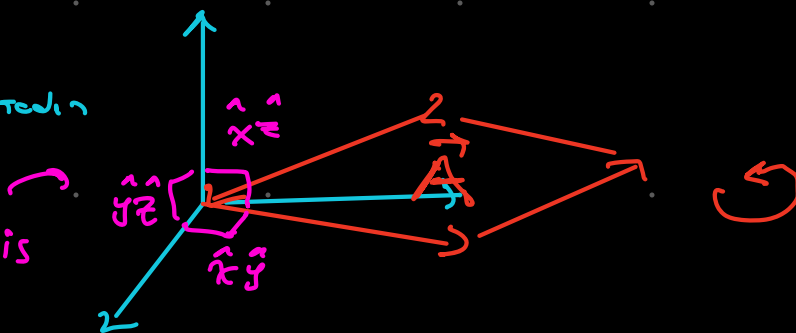
Vector



Bivector

(Area oriented)

bivector basis



$$\vec{A} = a \hat{x} \hat{y} + b \hat{y} \hat{z} + c \hat{x} \hat{z}$$

Trivector

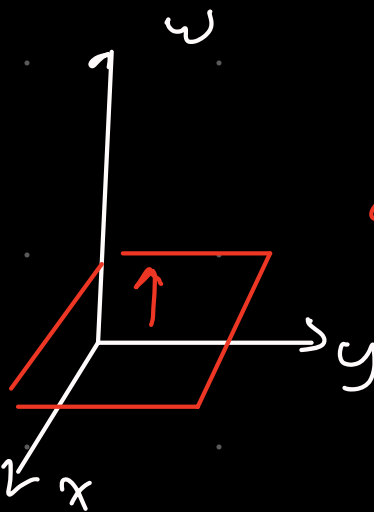
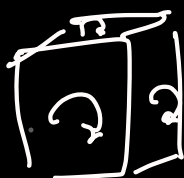
(Volume oriented)

in 3D

So Trivector

$$a \hat{x} \hat{y} \hat{z}$$

→ Pseudoscalar



Advantage of
Pseudoscalar
is that it
is a scalar
in 3D

Products

Product Interno

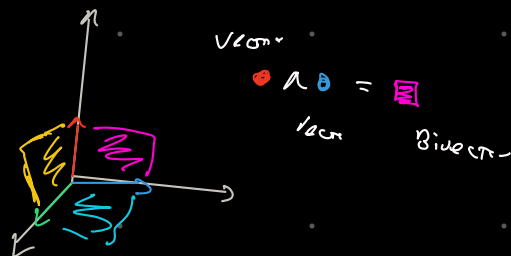
$$\vec{u} \cdot \vec{v} = \text{Scalar}$$

Product Externo (Gen) $\rightarrow$ anticomutativo

$$\vec{u} \wedge \vec{v} = \vec{A} \rightarrow \text{Bivector}$$

$$\vec{\text{Bivector}} \wedge \vec{\text{vector}} = \vec{\text{trivector}}$$

Generaliz-se ao produto



$$\vec{u} \vec{v} = \underbrace{\vec{u} \cdot \vec{v}}_{\text{scalar}} + \underbrace{\vec{u} \wedge \vec{v}}_{\text{Bivector}}$$

Exemplo

$$(3\hat{x} + 2\hat{y}) \cdot (6\hat{z} + 2\hat{x})$$

$$18\hat{x}\hat{z} + 6\hat{x}\hat{x} + 12\hat{y}\hat{z} + 4\hat{y}\hat{x}$$

$$\underbrace{6}_{\text{Scalar}} + \underbrace{18\hat{x}\hat{z} + 12\hat{y}\hat{z} + 4\hat{y}\hat{x}}_{\text{Bivector}} = \checkmark_{\text{multivector}}$$

Example (1) $(5\hat{x} + 2\hat{y}) (6\hat{x}\hat{y} + 5\hat{x}\hat{z} + 7\hat{y}\hat{z})$

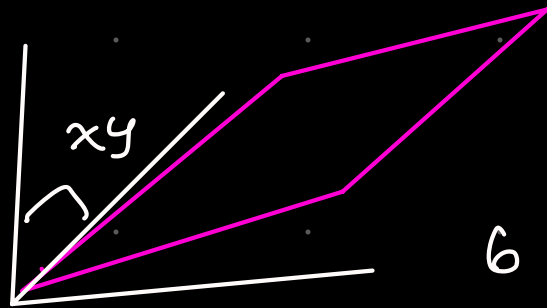
vector bivector

$$30\hat{x}\hat{x}\hat{y} + 25\hat{x}\hat{x}\hat{z} + 35\hat{x}\hat{y}\hat{z} + 12\hat{x}\hat{y}\hat{y} + 10\hat{y}\hat{x}\hat{z} + 14\hat{y}\hat{y}\hat{z}$$

$$30\hat{y} + 25\hat{z} + (35 - 10)\hat{x}\hat{y}\hat{z} + 12\hat{x}$$

trivector

$$12\hat{x} + 30\hat{y} + 25\hat{z} + 25\hat{x}\hat{y}\hat{z} = V_{\text{multivector}}$$



6 projects of the area
in xy

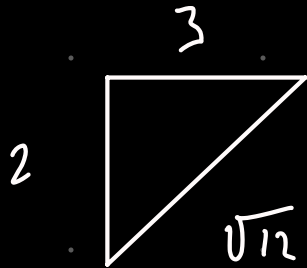
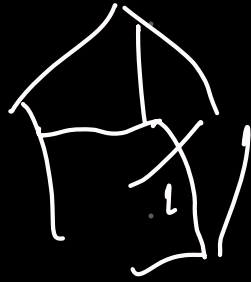
5 is the projection
on $\hat{x}\hat{z}$

7 is the projection
on $\hat{y}\hat{z}$

xy



$$xy + yz$$



Reviewing Maxwell's Equations

a look at the gradient

$$\vec{\nabla} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$$

the gradient is a vector

acting on scalar

$$\vec{\nabla}(\vec{E}) = \left(\frac{\partial E}{\partial x}, \frac{\partial E}{\partial y}, \frac{\partial E}{\partial z} \right)$$

$$\vec{\nabla} \cdot \vec{E} = \frac{\partial E}{\partial x} + \frac{\partial E}{\partial y} + \frac{\partial E}{\partial z}$$

$$\vec{\nabla} \times \vec{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix}$$

the gradient as vector

$$\vec{\nabla} = \frac{\partial}{\partial x} \hat{x} + \frac{\partial}{\partial y} \hat{y} + \frac{\partial}{\partial z} \hat{z}$$

$$\vec{E} = E_x \hat{x} + E_y \hat{y} + E_z \hat{z}$$

$$\vec{\nabla} \vec{E} = \underbrace{\frac{\partial E_x}{\partial x} \hat{x} \cdot \hat{x}}_{\text{Scalar}} + \underbrace{\frac{\partial E_y}{\partial x} \hat{x} \hat{y} + \frac{\partial E_x}{\partial y} \hat{y} \hat{x}}_{\text{Bivector}} + \underbrace{\frac{\partial E_y}{\partial y} \hat{y} \hat{y}}_{\text{Scalar}}$$

$$\vec{\nabla} \vec{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \hat{x} \hat{y} \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right)$$

$$\vec{\nabla} \vec{E} = \vec{\nabla} \cdot \vec{E} + \vec{\nabla} \vee \vec{E}$$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad \vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$



$$\vec{\nabla} \vee \vec{E} = -i \underbrace{\frac{\partial \vec{B}}{\partial t}}_{\text{Bivector}} \quad \vec{\nabla} \vee \vec{B} = i \mu_0 \underbrace{\left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)}_{\text{Bivector}}$$

$$\underbrace{\vec{\nabla} \cdot \vec{E}}_{\text{Bivector}} + \underbrace{\vec{\nabla} \vee \vec{E}}_{\text{Bivector}} + \underbrace{\vec{\nabla} \cdot \vec{B}}_{\text{Bivector}} + \underbrace{\vec{\nabla} \vee \vec{B}}_{\text{Bivector}}$$

$$\vec{\nabla} \left(\vec{E} + \vec{B} \right) = -\frac{i \partial \vec{B}}{\partial t} + i \mu_0 \epsilon_0 \left(\vec{J} + \frac{\partial \vec{E}}{\partial t} \right) + \frac{\rho}{\epsilon_0}$$

$$i \rho_0 \epsilon_0 \left(\frac{\vec{J}}{\epsilon_0} + \frac{\partial \vec{E}}{\partial t} - \frac{1}{\epsilon_0 \mu_0} \frac{\partial \vec{B}}{\partial t} \right)$$

Bivector

$$\frac{i \vec{J}}{c^2 \epsilon_0} + \frac{\rho}{\epsilon_0} + \frac{i}{c^2} \frac{\partial \vec{E}}{\partial t} - i \frac{\partial \vec{B}}{\partial t}$$

$$\nabla(\vec{E} + \vec{B}) = \frac{i \vec{J} + c^2 \rho}{\epsilon_0} + i \left(\frac{1}{c^2} \frac{\partial \vec{E}}{\partial t} - \frac{\partial \vec{B}}{\partial t} \right)$$

$$\nabla(\vec{E} + \vec{B}) - i \left(\frac{1}{c^2} \frac{\partial \vec{E}}{\partial t} - \frac{\partial \vec{B}}{\partial t} \right) = \frac{i \vec{J} + c^2 \rho}{\epsilon_0}$$

$$\eta = \left(\frac{\partial}{\partial t} + \nabla \right)$$

Carlos Weber
1990

$$\nabla(\vec{E} + \vec{B}) - i \frac{\partial}{\partial t} \left(\frac{\vec{E}}{c^2} - \vec{B} \right) = \frac{i \vec{J} + c^2 \rho}{\epsilon_0}$$